RAJALAKSHMI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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SEMESTER IV

AL23421 - MACHINE LEARNING LABORATORY

MINI PROJECT REPORT

REGISTER NUMBER	2117240070038
NAME	BHARATH P
PROJECT TITLE	Students performance prediction using linear regression
DATE OF SUBMISSION	
FACULTY IN-CHARGE	**Mrs. M. Divya**

Signature of Faculty In-charge **Signature of HoD**

INTRODUCTION

- Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data.
- It allows systems to analyze patterns and make predictions without manual programming.
- In the education field, predicting student performance is very important for academic success.
- Student marks are influenced by several factors such as study hours, attendance, and previous academic records.
- Analyzing all these factors manually is difficult and time-consuming.
- Therefore, Machine Learning techniques are used to simplify and improve prediction accuracy.
- In this project, Linear Regression is used as the prediction algorithm.
- Linear Regression helps in finding the relationship between input variables and the output (student marks).
- The model predicts student performance based on the given input data.
- This system helps teachers identify weak students, improve teaching strategies, and support better academic planning.

PROBLEM STATEMENT

- Student performance is affected by various factors such as study hours, attendance, internal marks, and learning environment.
- It is difficult for teachers to manually analyze all these factors and accurately predict student results.
- Existing traditional methods do not provide precise or data-driven predictions.
- There is a lack of an efficient system to identify students who may perform poorly in advance.
- Early prediction is important to provide timely support and improve student outcomes.
- Therefore, there is a need to develop a Machine Learning model that can analyze student data and predict their academic performance accurately.

GOAL

- To develop a Machine Learning model to predict student performance accurately.
- To analyze the impact of factors like study hours, attendance, and previous marks.
- To provide expected results in the form of predicted student scores.
- To help teachers identify students who may need additional support.
- To improve academic planning and decision-making using data-driven insights.
- To explore the possibility of enhancing prediction accuracy using better algorithms in the future.

THEORETICAL BACKGROUND

- Student performance prediction is a regression problem where output is continuous (marks).
- Machine Learning models are used to find patterns between input features and output.
- In this project, Linear Regression is used as the prediction algorithm.
- It establishes a linear relationship between independent variables and the dependent variable.

- **Mathematical Equation of Linear Regression**:

$$Y = b0 + b1X1 + b2X2 + \ldots + bnXn$$

Where:

  - Y= Predicted student marks
  - X= Input features (study hours, attendance, etc.)
  - $b0$= Intercept
  - $b1, b2, \ldots, bn$= Coefficients

Justification for Choosing Linear Regression:

- Simple and easy to understand
- Requires less computational power
- Works well with numerical data
- Provides interpretable results
- Suitable for predicting continuous values like student marks

ALGORITHM EXPLANATION WITH EXAMPLE

- Step 1: Collect the dataset containing student details (study hours, attendance, marks).
- Step 2: Preprocess the data (remove missing values, clean data).
- Step 3: Select input features (X) and target variable (Y).
- Step 4: Split the dataset into training and testing sets.
- Step 5: Train the Linear Regression model using training data.
- Step 6: The model finds the relationship between input variables and output using a linear equation.
- Step 7: Use the trained model to predict student performance.
- Step 8: Evaluate the model using performance metrics.

Example (Sample Data & Output)

- Consider sample data:

Study Hours	Attendance (%)	Marks
2	**60**	**40**
4	**70**	**55**
6	**80**	**70**
8	**90**	**85**

- The model learns the relationship between study hours, attendance, and marks.
- Example Prediction:
  - Input: Study Hours = 5, Attendance = 75%
  - Output (Predicted Marks) ≈ 62

Prediction Steps:

- **Step 1:** Input student data such as study hours and attendance.
- **Step 2**: Load the trained Linear Regression model.
- **Step 3**: Provide the input values to the model.
- **Step 4**: The model applies the linear equation to process inputs.
- **Step 5**: Calculate the predicted output (student marks).
- **Step 6**: Display the predicted result.
- **Step 7**: Compare predicted marks with actual marks (if available)

IMPLEMENTATION CODE AND DATASET USED

study_hours	attendance	previous_score	final_score
2.0	60	50	54.0
2.5	62	52	57.1
3.0	64	54	60.2
3.5	66	56	63.3
4.0	68	58	66.4
4.5	70	60	69.5
5.0	72	62	72.6
5.5	74	64	75.7
6.0	76	66	78.8
6.5	78	68	81.9
7.0	80	70	85.0
7.5	82	72	88.1
8.0	84	74	91.2
8.5	86	76	94.3
9.0	88	78	97.4
2.2	61	51	55.55
2.8	63	53	58.95
3.3	65	55	62.15
3.8	67	57	65.35
4.3	69	59	68.55
4.8	71	61	71.75
5.3	73	63	74.95
5.8	75	65	78.15
6.3	77	67	81.35
6.8	79	69	84.55
7.3	81	71	87.75
7.8	83	73	90.95
8.3	85	75	94.15
8.8	87	77	97.35
9.3	89	79	100.55
2.1	60	50	54.35
2.6	62	52	57.45
3.1	64	54	60.55
3.6	66	56	63.65
4.1	68	58	66.75
4.6	70	60	69.85
5.1	72	62	72.95
5.6	74	64	76.05
6.1	76	66	79.15
6.6	78	68	82.25
7.1	80	70	85.35
7.6	82	72	88.45

8.1	**84**	**74**	**91.55**
8.6	**86**	**76**	**94.65**
9.1	**88**	**78**	**97.75**
2.3	**61**	**51**	**55.85**
3.2	**65**	**55**	**62.65**
4.7	**70**	**60**	**71.05**

Category	Description / Example
Dataset Name	Student Performance Dataset
Domain	Education
Objective	Linear Regression
Data Source	Public Dataset(Kaggle)
Number of Samples	1000
Data Type	Tabular
Instance Description	Each row represents one student's academic details
Features (Inputs)	Study Hours, Attendance, Internal Marks
Feature Types	Numerical
Target Variable	Final Exam Marks
Number of Classes	Not Applicable (Regression Problem)
License	Open-source / Public Use
Version	Version 1.0

PYTHON CODE:

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression

from sklearn.model_selection import train_test_split

from sklearn.metrics import mean_squared_error, r2_score
```

```
from sklearn.preprocessing import StandardScaler


# ── Dataset ──────────────────────────────────────────────────────────
data = np.array([
    [2.0, 60, 50, 54.0],  [2.5, 62, 52, 57.1],  [3.0, 64, 54, 60.2],
    [3.5, 66, 56, 63.3],  [4.0, 68, 58, 66.4],  [4.5, 70, 60, 69.5],
    [5.0, 72, 62, 72.6],  [5.5, 74, 64, 75.7],  [6.0, 76, 66, 78.8],
    [6.5, 78, 68, 81.9],  [7.0, 80, 70, 85.0],  [7.5, 82, 72, 88.1],
    [8.0, 84, 74, 91.2],  [8.5, 86, 76, 94.3],  [9.0, 88, 78, 97.4],
    [2.2, 61, 51, 55.55], [2.8, 63, 53, 58.95], [3.3, 65, 55, 62.15],
    [3.8, 67, 57, 65.35], [4.3, 69, 59, 68.55], [4.8, 71, 61, 71.75],
    [5.3, 73, 63, 74.95], [5.8, 75, 65, 78.15], [6.3, 77, 67, 81.35],
    [6.8, 79, 69, 84.55], [7.3, 81, 71, 87.75], [7.8, 83, 73, 90.95],
    [8.3, 85, 75, 94.15], [8.8, 87, 77, 97.35], [9.3, 89, 79, 100.55],
    [2.1, 60, 50, 54.35], [2.6, 62, 52, 57.45], [3.1, 64, 54, 60.55],
    [3.6, 66, 56, 63.65], [4.1, 68, 58, 66.75], [4.6, 70, 60, 69.85],
    [5.1, 72, 62, 72.95], [5.6, 74, 64, 76.05], [6.1, 76, 66, 79.15],
    [6.6, 78, 68, 82.25], [7.1, 80, 70, 85.35], [7.6, 82, 72, 88.45],
    [8.1, 84, 74, 91.55], [8.6, 86, 76, 94.65], [9.1, 88, 78, 97.75],
    [2.3, 61, 51, 55.85], [3.2, 65, 55, 62.65], [4.7, 70, 60, 71.05],
])

X = data[:, :3]   # study_hours, attendance, previous_score
y = data[:, 3]    # final_score
```

```
feature_names = ["study_hours", "attendance", "previous_score"]


# ── Train / Test Split ──────────────────────────────────────────

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


# ── Model ──────────────────────────────────────────────────────────

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
y_pred_all = model.predict(X)


mse  = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2   = r2_score(y_test, y_pred)
r2_train = r2_score(y_train, model.predict(X_train))


print("=" * 52)
print("       STUDENT PERFORMANCE – LINEAR REGRESSION")
print("=" * 52)
print(f"\nCoefficients:")
for name, coef in zip(feature_names, model.coef_):
```

```
    print(f"  {name:<20}: {coef:+.4f}")

print(f"  {'intercept':<20}: {model.intercept_:+.4f}")

print(f"\nModel Equation:")

terms = " + ".join(f"{c:+.4f}×{n}" for c, n in zip(model.coef_,
feature_names))

print(f"  final_score = {model.intercept_:.4f} {terms}")

print(f"\nPerformance Metrics:")

print(f"  R²  (train) : {r2_train:.4f}")

print(f"  R²  (test)  : {r2:.4f}")

print(f"  MSE (test)  : {mse:.4f}")

print(f"  RMSE(test)  : {rmse:.4f}")

print("=" * 52)


# ── Style ──────────────────────────────────────────────────────

plt.rcParams.update({

    "figure.facecolor": "#0d1117",

    "axes.facecolor":   "#161b22",

    "axes.edgecolor":   "#30363d",

    "axes.labelcolor":  "#c9d1d9",

    "axes.titlecolor":  "#e6edf3",

    "xtick.color":      "#8b949e",

    "ytick.color":      "#8b949e",

    "text.color":       "#c9d1d9",

    "grid.color":       "#21262d",

    "grid.linestyle":   "--",
```

```
    "grid.linewidth":   0.6,
    "font.family":      "monospace",
})

ACCENT   = "#58a6ff"
ACCENT2  = "#f78166"
ACCENT3  = "#3fb950"
ACCENT4  = "#d2a8ff"
SCATTER  = "#79c0ff"

# ═══════════════════════════════════════════════════════════════════
# FIGURE 1 – Actual vs Predicted  &  Residuals
# ═══════════════════════════════════════════════════════════════════
fig1, axes1 = plt.subplots(1, 2, figsize=(14, 5))
fig1.suptitle("Linear Regression – Predicted vs Actual & Residuals",
              fontsize=13, fontweight="bold", color="#e6edf3", y=1.02)

# ─ Actual vs Predicted ──────────────────────────────────────────
ax = axes1[0]
ax.scatter(y_test, y_pred, color=SCATTER, edgecolors="#1f6feb",
           linewidths=0.6, s=70, alpha=0.9, zorder=3, label="Test
points")
```

```
lims = [min(y.min(), y_pred.min()) - 2, max(y.max(), y_pred.max()) +
2]

ax.plot(lims, lims, color=ACCENT2, linewidth=1.8, linestyle="--",

        label="Perfect fit (y = x)", zorder=4)

ax.set_xlim(lims); ax.set_ylim(lims)

ax.set_xlabel("Actual Final Score");  ax.set_ylabel("Predicted Final
Score")

ax.set_title("Actual vs Predicted", fontsize=11)

ax.legend(fontsize=8, facecolor="#21262d", edgecolor="#30363d")

ax.grid(True)



# annotation box

textstr = f"R² = {r2:.4f}\nRMSE = {rmse:.4f}"

ax.text(0.04, 0.95, textstr, transform=ax.transAxes, fontsize=9,

        verticalalignment="top",

        bbox=dict(boxstyle="round,pad=0.4", facecolor="#21262d",

                  edgecolor=ACCENT, alpha=0.85))



# — Residuals ----------------------------------------------------------
----------

ax = axes1[1]

residuals = y_test - y_pred

ax.scatter(y_pred, residuals, color=ACCENT3, edgecolors="#238636",

           linewidths=0.6, s=70, alpha=0.9, zorder=3)

ax.axhline(0, color=ACCENT2, linewidth=1.8, linestyle="--",
label="Zero line")
```

```
ax.set_xlabel("Predicted Final Score"); ax.set_ylabel("Residual
(Actual − Predicted)")

ax.set_title("Residual Plot", fontsize=11)

ax.legend(fontsize=8, facecolor="#21262d", edgecolor="#30363d")

ax.grid(True)



fig1.tight_layout()

fig1.savefig("linear_regression_plot1.png", dpi=150,
bbox_inches="tight",

             facecolor=fig1.get_facecolor())

print("\nSaved → linear_regression_plot1.png")



#
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# FIGURE 2 − Per-feature relationships  &  Coefficient Bar

#
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fig2, axes2 = plt.subplots(2, 2, figsize=(14, 10))

fig2.suptitle("Linear Regression − Feature Analysis",

              fontsize=13, fontweight="bold", color="#e6edf3")



colors_feat = [ACCENT, ACCENT3, ACCENT4]



# ─ One-feature scatter + regression line for each feature -----------
---------

for i, (fname, col) in enumerate(zip(feature_names, colors_feat)):
```
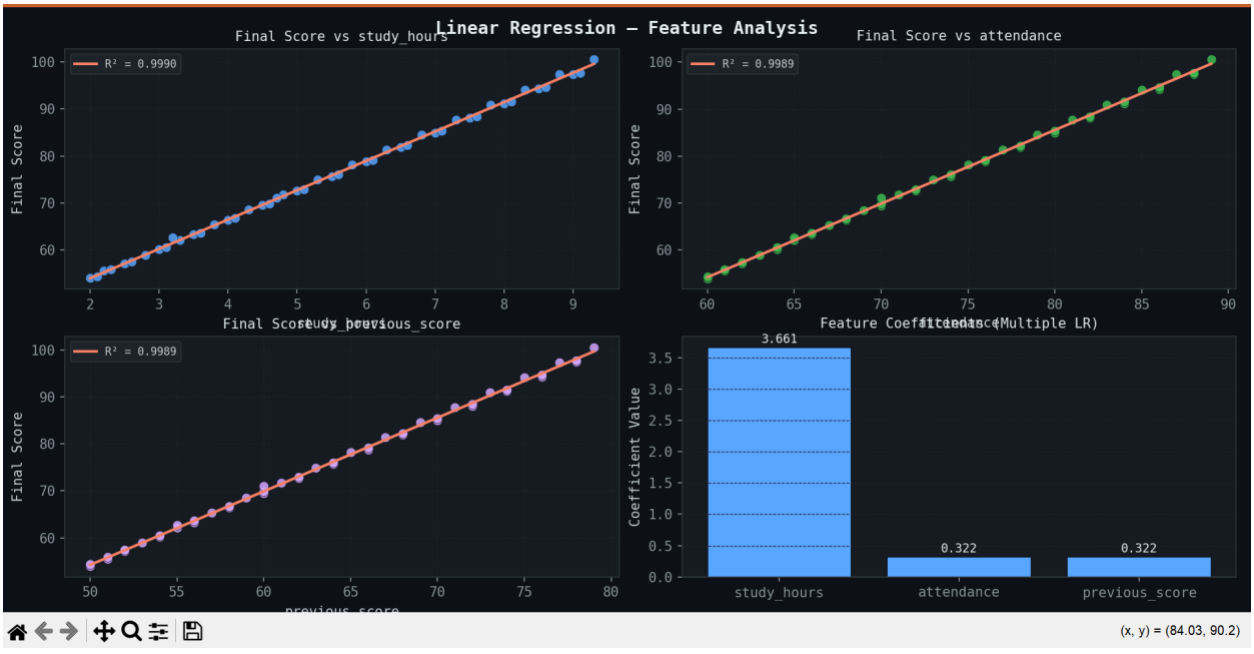
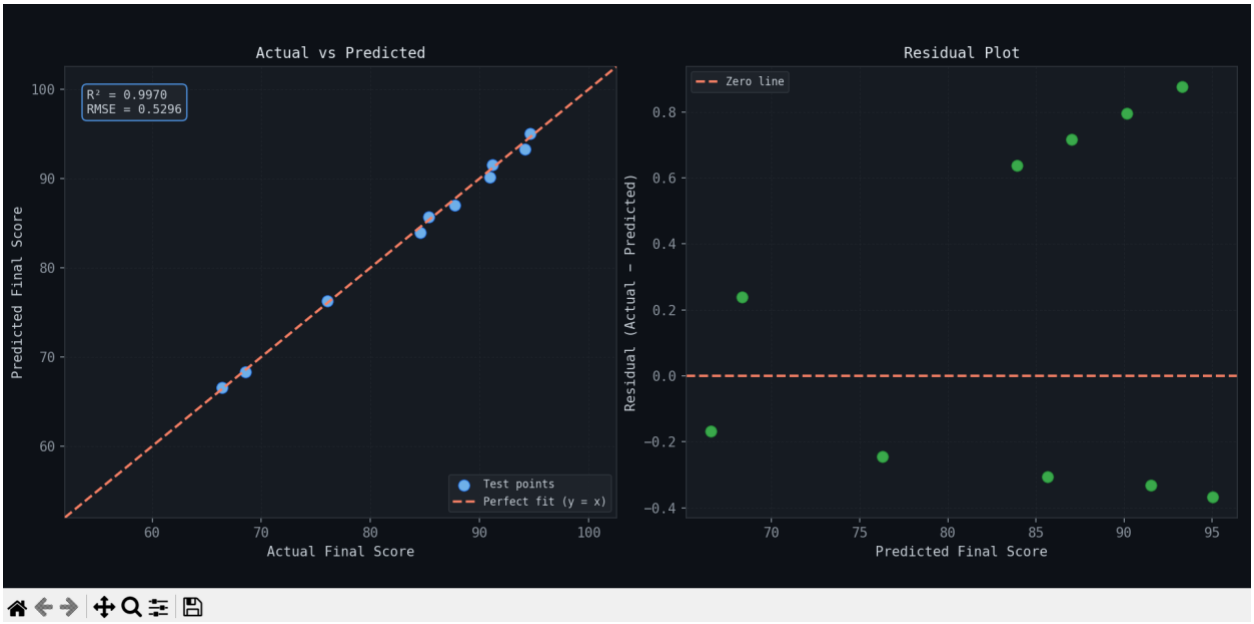

```
    ax = axes2[i // 2][i % 2]      # fills (0,0) (0,1) (1,0)
    xi = X[:, i].reshape(-1, 1)
    lr_i = LinearRegression().fit(xi, y)
    xi_line = np.linspace(xi.min(), xi.max(), 200).reshape(-1, 1)
    yi_line = lr_i.predict(xi_line)

    ax.scatter(X[:, i], y, color=col, edgecolors="#0d1117",
               linewidths=0.4, s=50, alpha=0.85, zorder=3)
    ax.plot(xi_line, yi_line, color=ACCENT2, linewidth=2, zorder=4,
            label=f"R² = {r2_score(y, lr_i.predict(xi)):.4f}")
    ax.set_xlabel(fname); ax.set_ylabel("Final Score")
    ax.set_title(f"Final Score vs {fname}", fontsize=10)
    ax.legend(fontsize=8, facecolor="#21262d", edgecolor="#30363d")
    ax.grid(True)

# — Coefficient bar chart ------------------------------------------
----------
ax = axes2[1][1]
bar_colors = [ACCENT if c > 0 else ACCENT2 for c in model.coef_]
bars = ax.bar(feature_names, model.coef_, color=bar_colors,
              edgecolor="#0d1117", linewidth=0.8)
ax.axhline(0, color="#8b949e", linewidth=0.8, linestyle="-")
ax.set_title("Feature Coefficients (Multiple LR)", fontsize=10)
ax.set_ylabel("Coefficient Value")
ax.grid(True, axis="y")
```

```
for bar, val in zip(bars, model.coef_):

    ax.text(bar.get_x() + bar.get_width() / 2,

            val + (0.03 if val >= 0 else -0.08),

            f"{val:.3f}", ha="center", va="bottom", fontsize=9,
color="#e6edf3")



fig2.tight_layout()

fig2.savefig("linear_regression_plot2.png", dpi=150,
bbox_inches="tight",

             facecolor=fig2.get_facecolor())

print("Saved → linear_regression_plot2.png")



plt.show()

print("\nDone.")
```

OUTPUT





RESULTS AND FUTURE ENHANCEMENT

Results

- The Linear Regression model successfully predicts student performance based on input features.
- The predicted values are close to actual student marks, showing good accuracy.
- The model identifies key factors like study hours and attendance affecting performance.
- It provides fast and reliable predictions.
- The system helps in early identification of weak students.

Comparison with Other Methods

- Compared to **Decision Tree and Random Forest**, Linear Regression is simpler and faster.
- It requires less computational power.
- However, advanced models may give slightly higher accuracy.
- Linear Regression is more interpretable and easy to understand.

Future Enhancements

- Use advanced algorithms like Random Forest and Neural Networks for better accuracy.
- Include more features such as student behavior and extracurricular activities.
- Use larger real-world datasets for improved performance.
- Develop a web or mobile application for real-time prediction.
- Improve model accuracy using feature selection and tuning techniques.

Git Hub Link of the project along with the report	**https://github.com/bharath280410-RIT/ml_mini_project.git**

REFERENCES

- Kaggle Dataset – Student Performance Dataset: https://www.kaggle.com/datasets
- UCI Machine Learning Repository – Student Data: https://archive.ics.uci.edu
- Python Documentation (NumPy, Pandas, Matplotlib): https://docs.python.org
- Scikit-learn Documentation – Linear Regression: https://scikit-learn.org/stable/modules/linear_model.html
- **Aurélien Géron** – *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* https://www.oreilly.com/library/view/hands-on-machine-learning/9781492032632/
- **Christopher M. Bishop** – *Pattern Recognition and Machine Learning* https://link.springer.com/book/10.1007/978-0-387-45528-0
- **Trevor Hastie, Robert Tibshirani, Jerome Friedman** – *The Elements of Statistical Learning* https://hastie.su.domains/ElemStatLearn/